

Nikshiptha Sonajoke

Computer Science & Engineering Undergraduate | IIT Jodhpur
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EDUCATION

IIT JODHPUR

B.TECH IN COMPUTER SCIENCE & ENGINEERING

CGPA : 6.20/10

2022-2026 | Jodhpur, India

SRI CHAITANYA COLLEGE

CLASS 12 | TSBIE

MARKS : 986/1000

2022 | Hyderabad, Telangana

SRI CHAITANYA SCHOOL

CLASS 10 | BSET

CGPA : 10

2020 | Hyderabad, Telangana

COURSEWORK

UNDERGRADUATE

Data Structures & Algorithms

Operating Systems

Database Management Systems

Computer Networks

Software Engineering

Computer Architecture

Cyber Security

Cloud Computing

Machine Learning

SKILLS

PROGRAMMING LANGUAGES

C++ • Python • C • SQL/MySQL

ANALYTICS/ML

Data Analysis • Scikit-Learn • NumPy •

Pandas • OpenCV

BACKEND/WEB

Flask • REST APIs • HTML • CSS •

JavaScript

TOOLS & PLATFORMS

Git • Linux • Docker • VS Code

EXTRACURRICULAR

• **Core Member – Dramatics Society, IIT Jodhpur**

Participated in team-based stage productions and event activities.

• **Assistant Head – Prometeo'24, IIT Jodhpur**

Coordinated teams and supported event execution for the institute technical fest.

EXPERIENCE

INDIANOIL CORPORATION | MACHINE LEARNING INTERN

Guwahati Refinery

- Built a YOLOv8-based computer vision system for automated PPE compliance monitoring using refinery surveillance footage.
- Fine-tuned a custom detection model on 5,000+ annotated frames for real-time helmet detection and violation tracking.
- Implemented automated logging pipelines with timestamps, bounding boxes, and captured snapshots for audit and monitoring purposes.
- Developed automated monitoring and logging workflows to improve incident tracking and operational efficiency.
- Tools: Python, YOLOv8, OpenCV, Roboflow

PROJECTS

ADOBE INDIA HACKATHON 2025 – ETL/DATA PROCESSING PIPELINE | ETL/DATA PROCESSING PIPELINE |

Jul 2025 | Python, PyMuPDF, Docker

- Designed and implemented an automated data processing pipeline to extract and transform structured information from unstructured documents.
- Optimized parsing and extraction logic for large multi-page documents with improved processing efficiency.
- Processed and organized extracted document data into structured JSON outputs for scalable downstream processing workflows.
- Built a Dockerized solution for consistent execution and streamlined deployment workflows.

CROP RECOMMENDATION | ML + WEB DEVELOPMENT |

Feb--May 2024 | Flask, Python, Random Forest

- Developed a Flask-based web application to recommend crops using soil and environmental parameters.
- Built backend workflows for data preprocessing, validation, and model inference using Random Forest.
- Compared model performance with Logistic Regression and XGBoost for prediction evaluation.
- Developed frontend interfaces using HTML, CSS, and JavaScript for real-time user interaction.

STUDENT RESULT PORTAL | DBMS COURSE PROJECT |

Sep--Nov 2024 | PHP, MySQL, HTML, Bootstrap

- Developed a role-based portal for managing student academic records and result workflows.
- Implemented authentication and access control for students, faculty, and administrators.
- Designed structured MySQL database schemas for registration, result management, and reporting operations.
- Optimized SQL queries and structured relational schemas for efficient academic record management.
- Built functionalities for record updates, result uploads, and SQL-based report generation.